Unit 8, Lesson 8: Finding the Missing Dimension of a Rectangular Prism

Benchmark

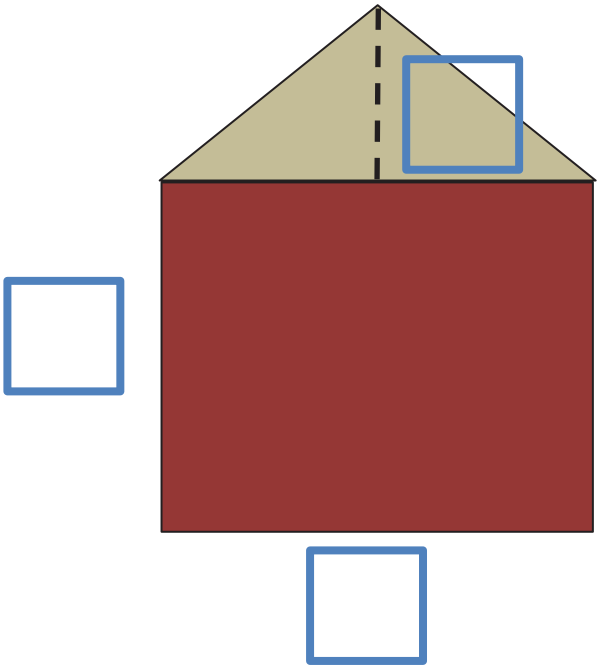
MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.

Learning Target

- ☐ I can find the value of a missing dimension of a rectangular prism.

8.8.1 Warm-Up: Area of a Wall

1. Using the digits $0-9$, at most one time each, fill in the boxes to make a true statement.



Area of  =

Area of  =

8.8.2 Collaboration: Missing Measurement

Jolene works for a textbook company filling orders. She has discovered each textbook has a width of 7 inches (in.), a height of $2\frac{2}{3}$ in., and a length of 10 in.

The textbook company has decided to make the textbooks smaller by reducing the length. The new volume of the textbook will be $168\ \text{in}^3$.

1. Complete the table.

Old Textbook	New Textbook
$V = l \times w \times h$ $V =$ ______ $\times$ ______ $\times$ ______ $V =$ ________ in^3	$V = l \times w \times h$ $168\ \text{in}^3 = l \times$ _____in. $\times$ _____in. $168\ \text{in}^3 =$ ______in. $\times$ ______in^2

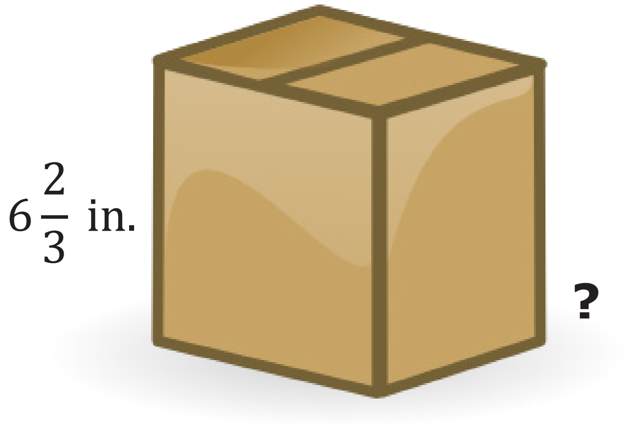
2. Work with your partner to determine the new length of the textbook. Explain your process.

8.8.3 Guided Instruction: Finding a Missing Dimension

Each flat surface of a prism is called a face.

When only two dimensions of a three-dimensional prism are given, the product of the two given dimensions represents the area of that face of the prism.

1. What is the area of the front face of this prism? Include units.



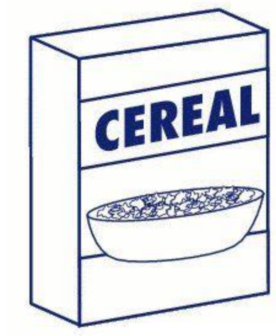
The volume of the prism is equal to the area of the face multiplied by the length of the missing dimension. The value of the missing dimension can then be calculated by dividing the volume by the area of the face.

$$V = Area\ of\ the\ face \times measurement\ of\ the\ missing\ dimension$$

$$Measurement\ of\ the\ missing\ dimension = \frac{V}{Area\ of\ the\ face}$$

2. If the volume of the box is $233\frac{1}{3}$ in^3, what is the value of the missing dimension of the box?

3. Rafael was trying to find the missing dimension of the cereal box shown.



30 cm

20.3 cm

Volume = $1{,}218 \text{ cm}^3$

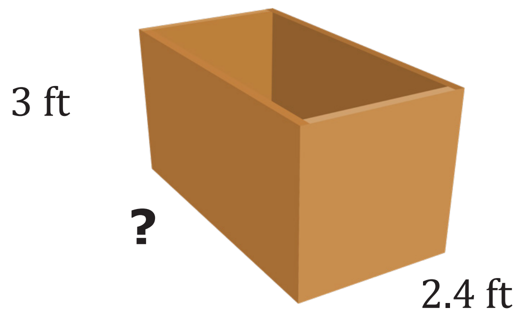
He made a mistake while solving for the width of the box, but he cannot find it. Inspect his work below.

Rafael's Work
20.3 cm × 30 cm 60.9 cm² Area of face 20 cm³ Volume 60.9) 12180 1218 0

Write a note to Rafael on how to correct his error(s).

8.8.4 Guided Practice: Find the Missing Dimension

1. Sally broke her ruler before she finished measuring her toy chest. She knows the toy chest has a volume of 25.2 ft^3.

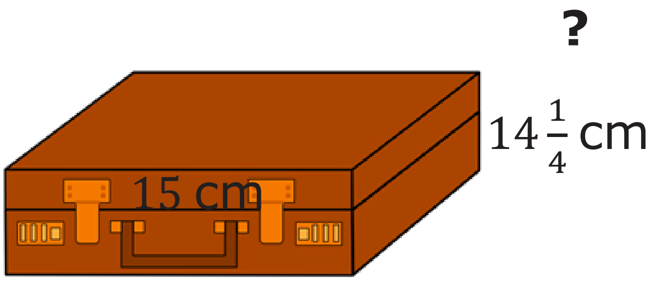


What is the length of the toy chest?

2. Damion works in construction and needs to cut a wooden beam to fill a cement hole. He knows that the beam has a width of 1.2 meters (m) and a length of 0.25 m. If Damion knows that the volume of the beam needs to be 1.5 cubic meters (m^3), what is the height?

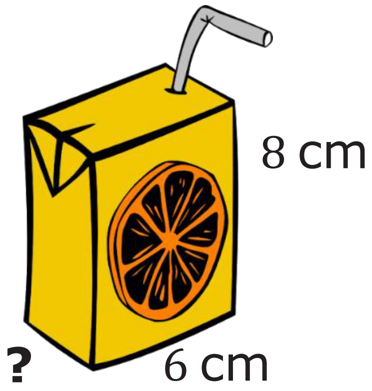
8.8.5 Your Turn!

1. Mr. Rodrigo knows that his briefcase has a volume of $1{,}068\frac{3}{4}$ cubic centimeters (cm^3). He labeled the dimensions of the briefcase that he knew.



What is the height of the briefcase?

2. Victor works in a factory that makes juice boxes. He knows that the juice boxes have a height of 8 centimeters (cm), a length of 6 cm, and a volume of 192 cubic centimeters (cm^3).



What is the width of the juice box?

8.8.6 Wrap-Up: Reflection

1. Rate your strength on the following areas using the scale.

1 – still learning
2 – making progress
3 – got it

Skill	Rating
Find the volume of a rectangular prism.	
Find a missing dimension of a rectangular prism.	

2. Write an explanation of why you rated yourself the way you did.